

Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Not only which carrier got bigger — which carrier moved the composition.

CoDaWork 2026 · Coimbra, Portugal · 1–5 June 2026

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Operationalizing compositional analysis — a runnable standard for researchers and the AI assistants they choose.

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share and large in structural work.

World electricity, 25 years — the standard stacked-area view

Coal stays dominant. Gas grows. Nuclear declines.
Solar appears as a thin yellow sliver after 2010.
Wind grows steadily but stays visually small until late.

What the size view misses

USA Solar, 2012 → 2013

starting share	0.107 %
structural Power Share	81.7 %
Activation Coefficient	$\approx 760 \times$

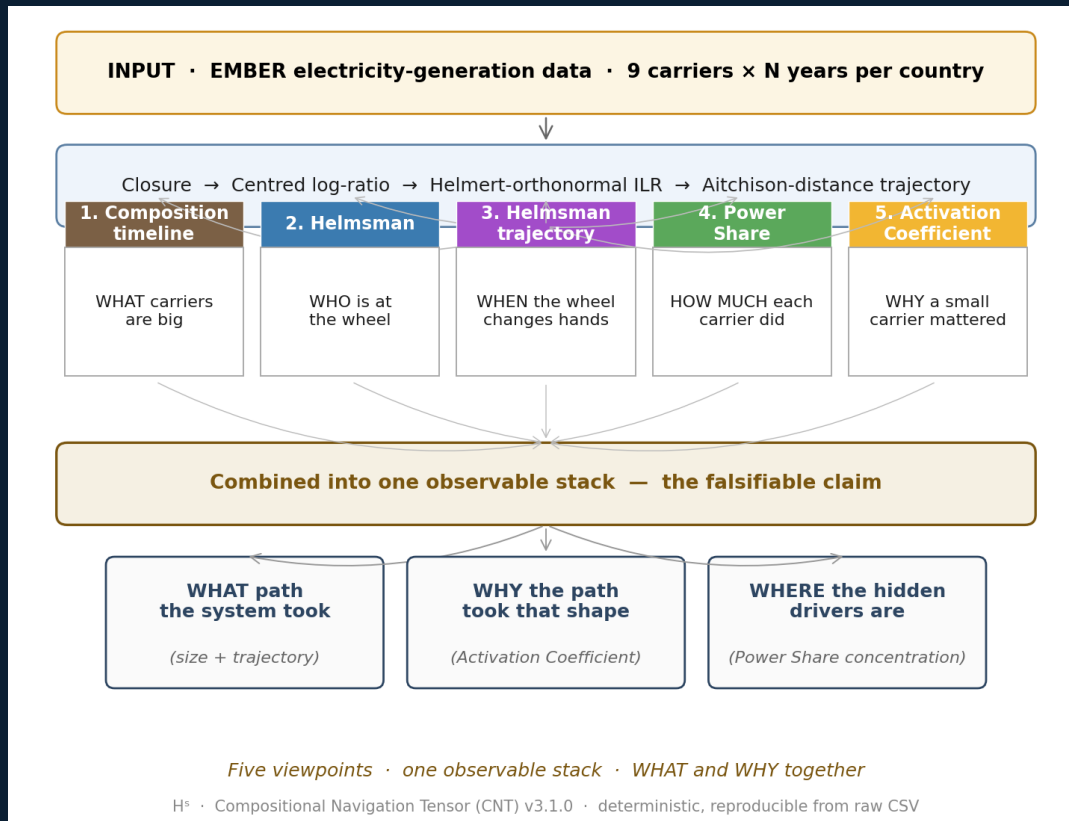
*Solar acted at $760 \times$ its size.
No size view shows that.*

This talk is the reason that number exists.

Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Five viewpoints, one observable stack

Each viewpoint answers one question. Together: an auditable transition event.



Composition

what share each carrier has.

Helmsman

which carrier has the largest CLR move at a step.

Helmsman trajectory

when the steering carrier changes.

Power Share

how much squared CLR motion each carrier did.

Activation Coefficient

Power Share ÷ starting share — the yeast factor.

All five derive from CLR + ILR-Helmert. Pure CoDa geometry; no new mathematics.

The Activation Coefficient — the yeast factor

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — USA Solar 2012 → 2013

starting share	0.107 %	small
Power Share	81.7 %	most of the work
α	$\approx 760 \times$	yeast

Yeast is 2% of a loaf by mass and does 100% of the rising. Same shape.

Solar 2010–2015 appears repeatedly as small-share / large-structural-work across the corpus.

Three archetypes — one instrument, three regimes

Same protocol applied to three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende
2000 → 2025
solar + wind absorb
structural work
before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011
displaces nuclear,
causes 2011–2013
multi-year compositional
reorganisation.

United Kingdom

regime change

jump and return

Coal exit 2012–2020
from > 30 % to < 2 %.
Wind, solar, others
absorb displaced
structural work.

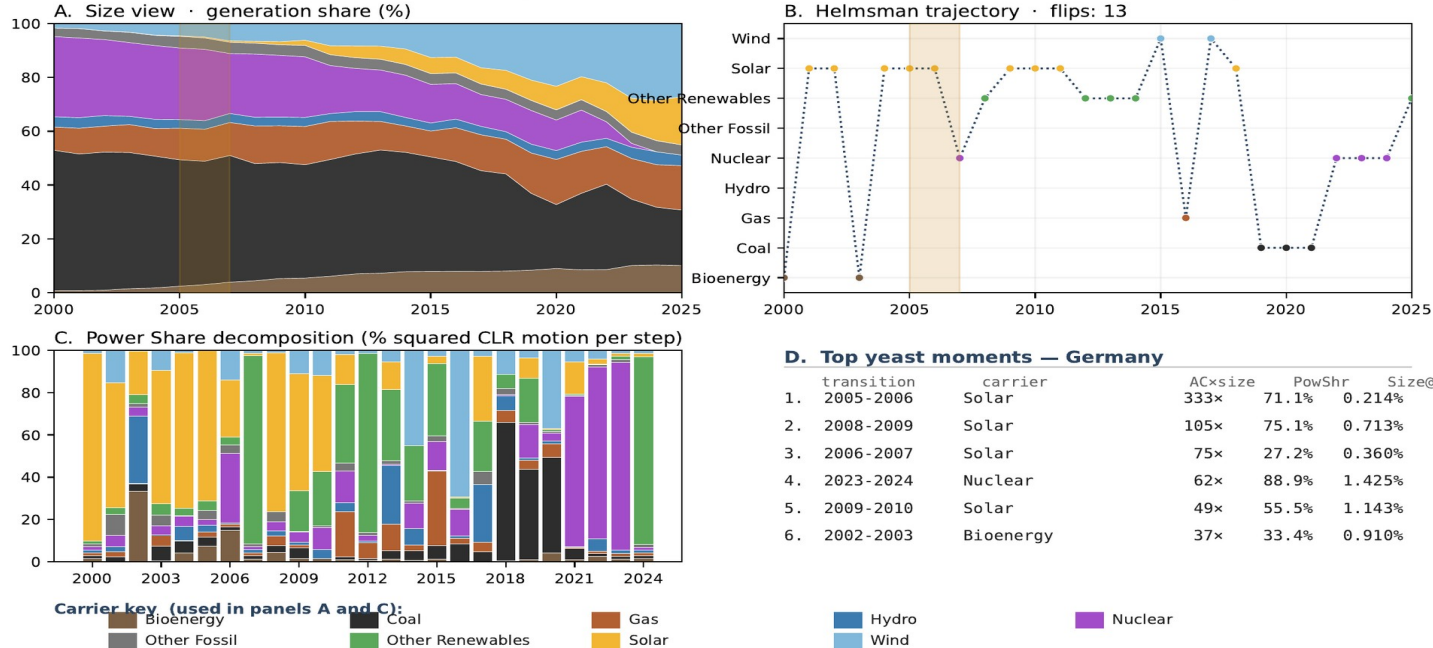
Three different transition regimes. One operational protocol reads them all.

Germany — deliberate transition as continuous course

The Energiewende read as a single smooth arc on the simplex.

Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The size view shift is real — the yeast moments show when it actually accelerated.



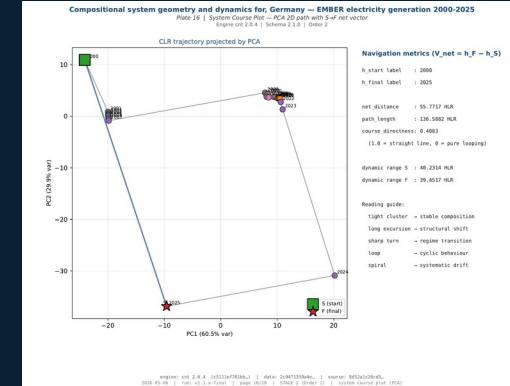
Germany's transition is the textbook case of a continuous arc: coal recedes, gas holds, solar and wind grow steadily.

But the Power Share view shows the acceleration moments — solar in 2005-2006 at 0.21 % composition share doing 71 % of the structural directional work (activation coefficient $\approx 333\times$).

That is the structural beginning of the Energiewende — three years before the size view would call it visible.

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Fig. 2



Solar 2005-2006: 0.21 % share · 71.1 % structural work · $\alpha \approx 333\times$

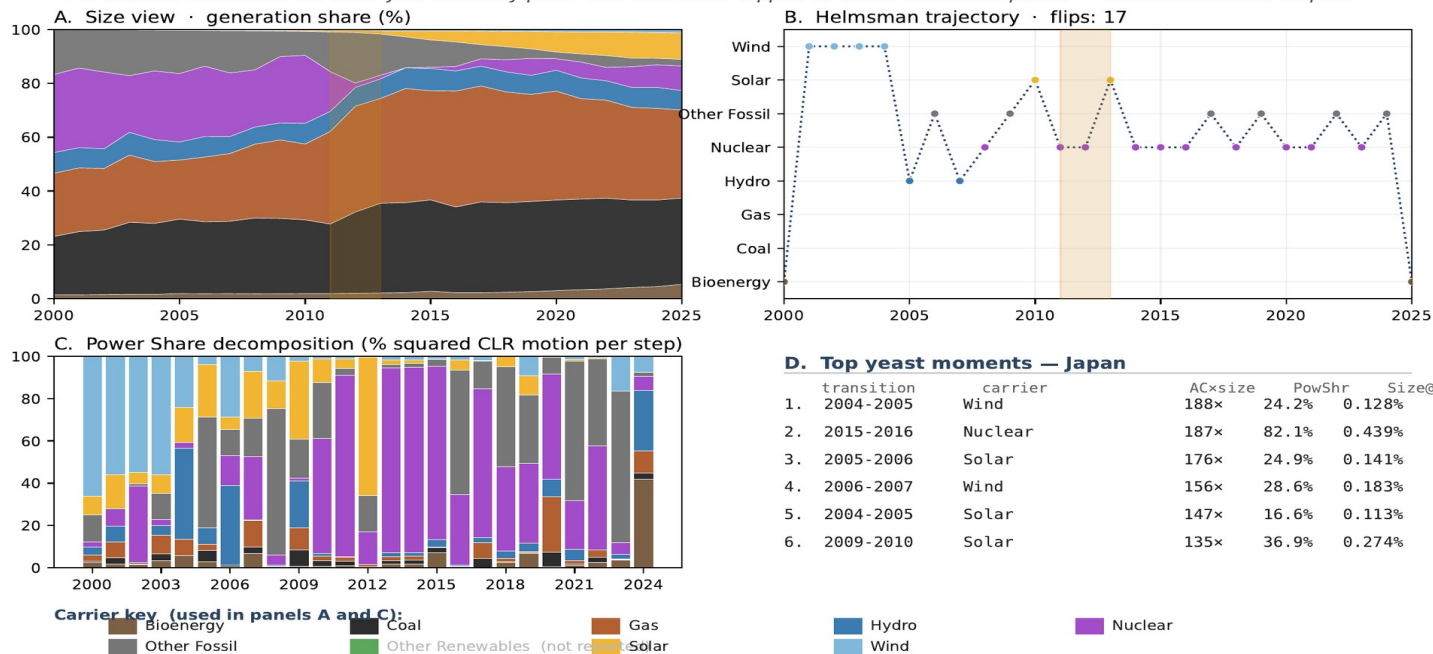
Helmsman trajectory: course directness 0.41 — continuous arc to renewable vertex.

Japan — Fukushima shock and reorganisation

The instrument detects both the shock and the multi-year reorganisation.

Japan — Fukushima 2011 and the structural aftermath

An external shock and the multi-year recovery path. The helmsman flipped 17 times across the period — the loudest in the corpus.

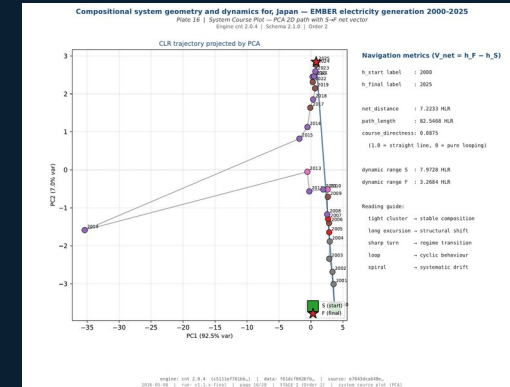


The 2011-2012 transition shows in every viewpoint at once: composition (nuclear collapses), helmsman (carriers shift abruptly), Power Share (massive concentration of squared CLR motion at one step).

The years AFTER tell the deeper story — Japan's post-Fukushima mix kept reorganising for over a decade, and the yeast moments distribute across multiple carriers as solar, gas, and renewables each took turns absorbing the displaced nuclear share.

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Fig. 3



Aitchison distance 2011 → 2012 ≈ 3 × neighbouring-year baseline · helmsman flips 17 ×

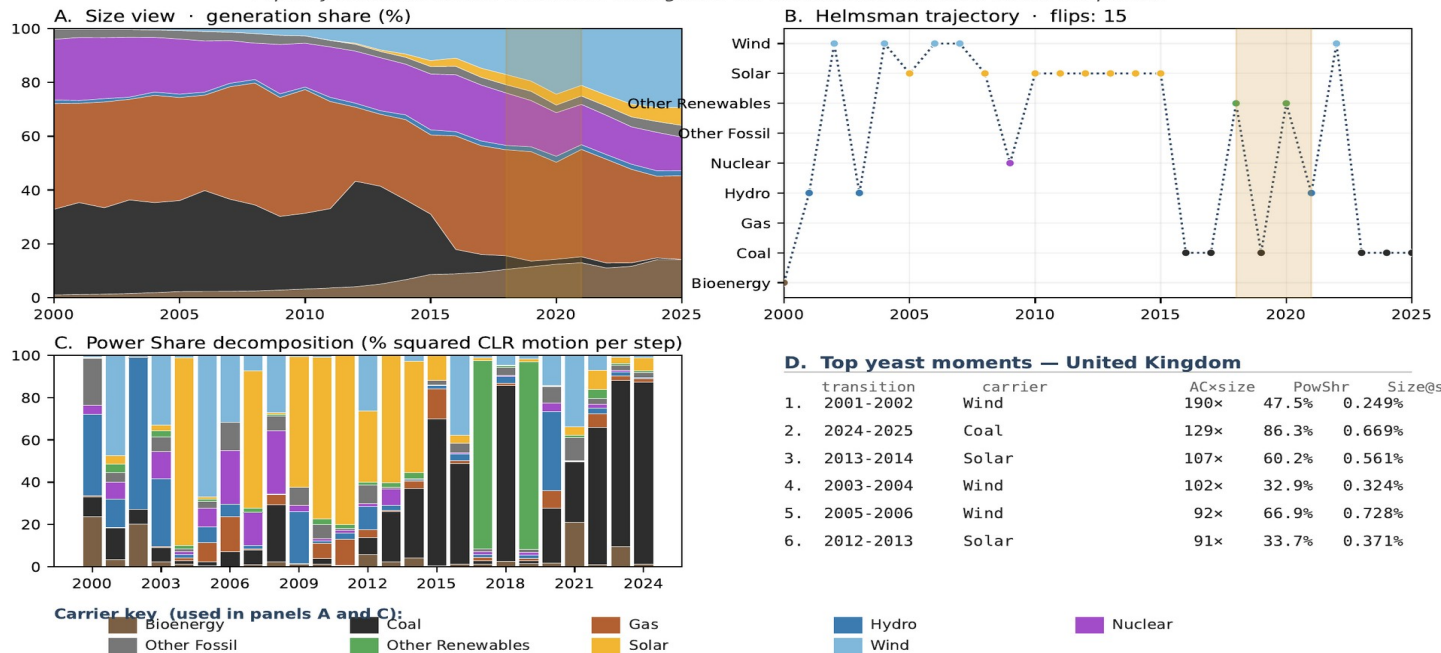
Course directness 0.09 — looping reorganisation, not a single step.

United Kingdom — coal exit as regime change

Policy-driven displacement, absorbed across multiple renewable carriers.

United Kingdom — the coal exit as a regime change

A policy-driven structural transition. Coal goes from dominant to near-zero across the period.



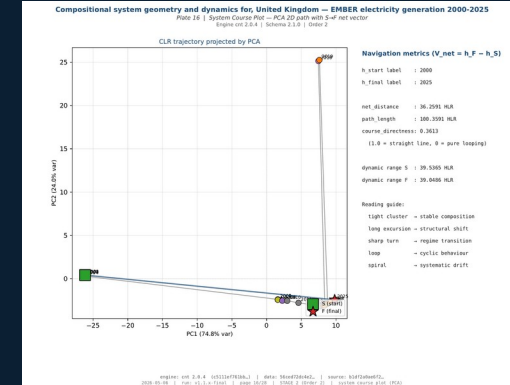
The UK story is the abrupt-policy archetype. Coal's share drops from over 30 % to under 2 % across the period.

The Power Share view names what made it possible: Other Renewables and Wind in the late-2010s took turns doing the structural work of absorbing coal's exit, often at low single-digit composition share (panel D).

Without the activation coefficient, those small-share carriers are invisible. With it, the whole transition snaps into focus.

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Fig. 4

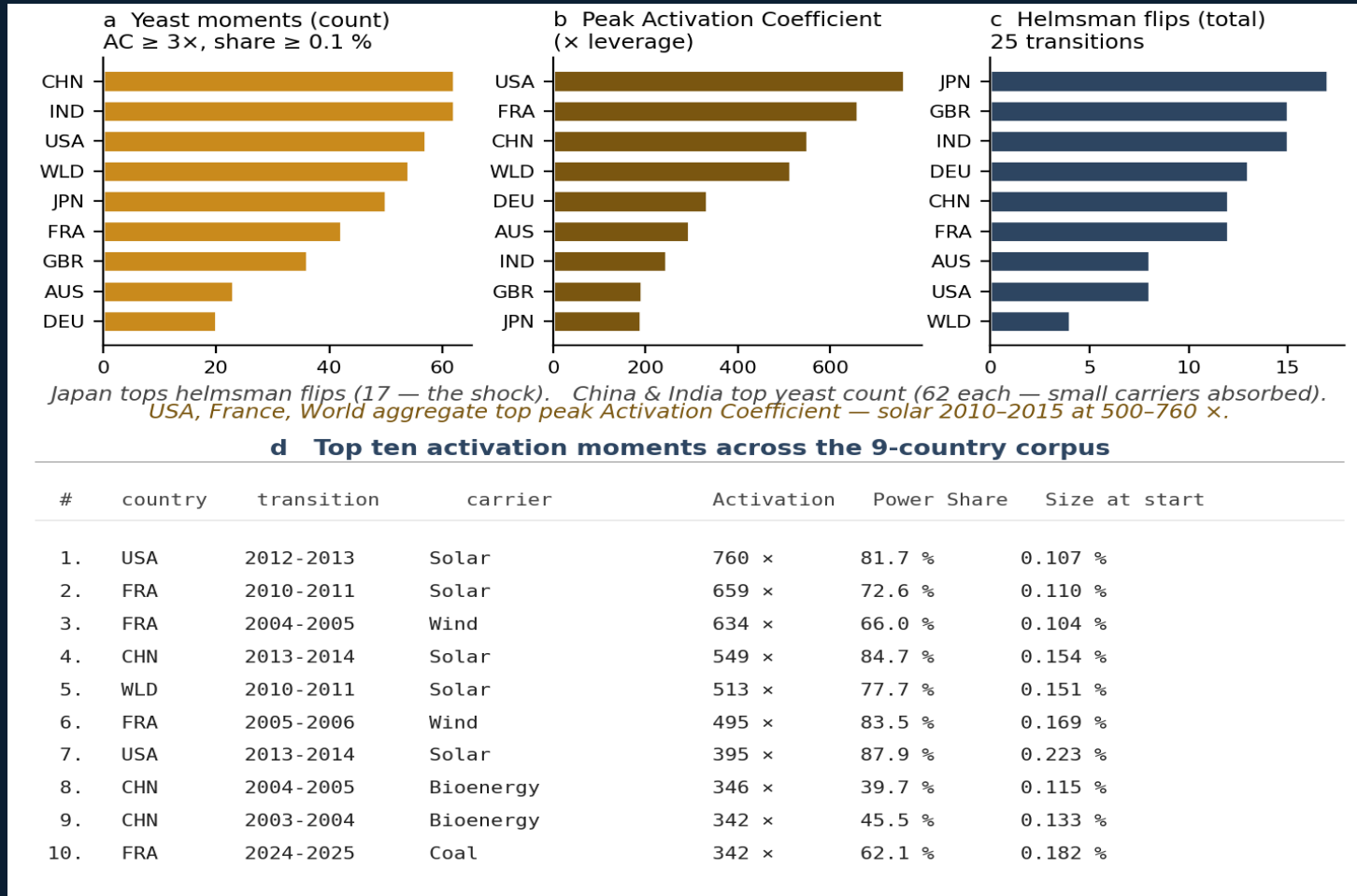


Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced structural work.

Course directness 0.36 — jump-and-return archetype.

Cross-country signature — 5 of 9 reproduce the pattern

From three case archetypes to a corpus-level result.



fires AUS • CHN • GBR • IND • JPN

quiet DEU (annual) • FRA • USA • WLD

A useful detector should not fire everywhere. Discrimination is itself evidence.

What the stack answers

One observable, five distinct questions, one reproducible object.

WHAT	carriers are big.	<i>size view</i>
WHO	is at the wheel.	<i>Helmsman</i>
WHEN	the steering changes.	<i>Helmsman trajectory</i>
HOW MUCH	work each carrier did.	<i>Power Share</i>
WHY	a small carrier mattered.	<i>Activation Coefficient</i>

The stack does not replace interpretation. It gives interpretation a reproducible object.

MC-4 — the falsifiable claim

Three conjuncts. Four defeat paths. One open invitation to the room.

The claim

H^s is the first published protocol that combines:

1. Aitchison-native compositional metrics
2. formal change-point / deceptive-drift detection
3. carrier-level structural-work attribution

Defeat paths (any one falsifies)

- A. prior-art path — show all three were already combined.
- B. metric path — show a defensible non-Aitchison metric.
- C. case path — show a country / year the protocol misreads.
- D. category path — show the construct itself is ill-posed.

This is the claim I am asking the CoDa community to test. Take it apart.

Now inspect the instrument

I will not ask you to trust the speaker. The outputs are open for inspection.

Manuscript

25-page paper,
Nature structure,
Supplementary Info,
six figures.

`papers/codawork2026/
manuscript/`

Cinema scroll

66-slide reel,
325-page master PDF.
Every plate the engine
produced for the
nine-country EMBER
corpus.
Scroll at speed.
Pause anywhere.

Projector

Offline HTML.
RADAR · BARY · ALIGN
+ SHOCK overlay.

Q&A backdrop.

No network call.
Runs in any browser.

[github.com / PeterHiggins19 / higgins-decomposition](https://github.com/PeterHiggins19/higgins-decomposition) · [CODA-Association/](#) · [QR on handout](#)

Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR). Operationalization standard, runnable.

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The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line. The AI follows the same protocol.